
Taming Aleatoric Impulse in Off-Policy Reinforcement Learning

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Abstract

Off-policy reinforcement learning suffers from the aleatoric impulse—a transient surge in value overestimation arising from intrinsic noise in empirical value estimation. Previous algorithms either overlook this issue or mitigate it through heuristic strategies, which inevitably introduce hyperparameter sensitivity and asymptotic inconsistency. To address this, we propose **Direct Evidential Regression (DEAR)**, a generalized operator that uses an exponential moving average (EMA) anchor to explicitly quantify the higher-order aleatoric uncertainty of arbitrary estimators in an end-to-end manner. Building upon DEAR, we develop **AlextbfAleatoric Impulse Damping via Direct Evidential Regression (AIDER)** which restructures the critic network to directly learn aleatoric value uncertainty. This principled uncertainty estimate eliminates the need for heuristic annealing schedules and naturally provides both a robust pessimistic target bound and a dynamic exploration bonus. Extensive experiments on MuJoCo, the DeepMind Control Suite, and a vehicle longitudinal tracking task demonstrate that AIDER achieves state-of-the-art performance on challenging dynamic benchmarks, consistently outperforming its predecessor DSAC-AID and other strong baselines.

1 Introduction

Deep reinforcement learning has achieved remarkable success across a wide range of complex sequential decision-making tasks [1, 2, 3, 4]. However, addressing the severe overestimation bias of action-values remains a fundamental challenge for stable off-policy optimization [5]. The root cause of this vulnerability lies in the intrinsic uncertainty associated with empirical value estimation. While modern ensemble-based algorithms have proven highly effective at mitigating epistemic uncertainty, which arises from structural model inaccuracies, they fundamentally struggle with the irreducible aleatoric uncertainty inherent in environmental and policy stochasticity [6]. During the early stages of training, the resonance of high policy stochasticity with severe data scarcity acutely amplifies this intrinsic noise. This amplification triggers a transient but destructive overestimation surge, defined as the *aleatoric impulse*, which directly misleads the agent toward suboptimal actions [7].

Despite advancements in distributional reinforcement learning, existing methods remain structurally vulnerable to this impulse. Approaches such as DSAC-T [8] and IV-RL [9] implicitly encapsulate the critical aleatoric uncertainty within the learned total return uncertainty, yet they fail to isolate it for target rectification. Instead, this information is confined solely to gradient modulation. We argue that such indirect mitigation, akin to network regularization in BRO [10], merely attenuates noise propagation but fails to rectify the inflated targets at their source. Conversely, while methods like DoubleGum [11] and DSAC-AID [7] attempt to enforce explicit penalization, they suffer from a fundamental structural misalignment. Specifically, they rely on heuristically expressing the critical aleatoric value uncertainty indirectly through the learned aleatoric return uncertainty. This indirect reconstruction inherently necessitates complex heuristic scaling, which introduces

severe hyperparameter sensitivity and asymptotic inconsistency, ultimately failing to provide a robust, generalized solution.

Fundamentally resolving these structural bottlenecks necessitates a paradigm shift toward directly estimating the value uncertainty, which is intrinsically a higher-order uncertainty. Evidential learning naturally aligns with this distinct property by modeling a higher-order distribution emerging as the ideal paradigm for the network to directly quantify its own estimation variance [12]. However, existing evidential learning frameworks are predominantly tailored for discrete spaces, such as classification tasks [13, 14]. While some recent extensions attempt to address continuous domains, their modeling mechanisms remain highly heuristic [15]. These continuous approaches are typically limited to rough out-of-distribution (OOD) detection and fundamentally lack the capacity to accurately estimate the precise higher-order aleatoric uncertainty required for target rectification [16, 17].

To transcend these limitations, we propose **Direct Evidential Regression (DEAR)**, a generalized target-anchored evidential learning paradigm. By utilizing an Exponential Moving Average (EMA) of the network weights to construct a stable empirical anchor, and formulating the optimization as a higher-order distribution matching problem, this universal operator leverages temporal consistency to learn the parameter estimation variance directly and end-to-end, completely bypassing heuristic approximations. By seamlessly embedding the DEAR operator into DSAC-AID, we further propose **AIDER (Aleatoric Impulse Damping via Direct Evidential Regression)** mechanism. Within this fully integrated framework, the directly learned aleatoric uncertainty is directly added to the epistemic uncertainty to obtain the total value uncertainty. This total uncertainty is then utilized for overestimation penalization and exploration encouragement, completely eliminating the reliance on complex heuristic annealing.

The main contributions of this work are twofold: (1) We propose a generalized **DEAR** operator. By bypassing the mathematical rigidity of traditional conjugate priors and utilizing EMA-driven empirical anchors, this paradigm provides a universal, end-to-end solution for quantifying the higher-order aleatoric uncertainty of arbitrary parametric estimators. (2) Building upon this generalized operator, we develop the **AIDER** mechanism. By directly learning the value uncertainty, AIDER gracefully tames the aleatoric impulse, achieving state-of-the-art robust evaluation and efficient exploration without any hyperparameter sensitivity or heuristic annealing schedules.

2 Preliminaries

2.1 Problem Formulation

We consider a standard reinforcement learning setting formulated as a Markov Decision Process (MDP) $(\mathcal{S}, \mathcal{A}, p, r, \gamma)$, where $s \in \mathcal{S}$ and $a \in \mathcal{A}$ represent the state and action, respectively [18]. The dynamics are governed by the transition probability $p(s'|s, a)$ and the reward function $r(s, a)$. The agent aims to learn a policy $\pi(a|s)$ that maximizes the maximum entropy objective: $J(\pi) = \mathbb{E}[\sum_{t=0}^{\infty} \gamma^t (r_t + \alpha \mathcal{H}(\pi(\cdot|s_t)))]$, where α is the temperature parameter controlling the stochasticity of the optimal policy.

2.2 Aleatoric Impulse

In off-policy reinforcement learning, the true action-value is inherently deterministic, defined as the theoretical expectation of the return distribution: $\tilde{Q}(s, a) = \mathbb{E}[Z(s, a)]$. In practice, however, algorithms approximate this true value using an empirical estimator $\hat{Q}$, parameterized by an ensemble of m finite neural networks trained on n effective samples:

$$\hat{Q}(s, a) \approx \frac{1}{m \cdot n(s, a)} \sum_{i=1}^m \sum_{j=1}^n z_{ij}, \text{ where } z_{ij} \sim Z_{\theta_i}(s, a). \quad (1)$$

Here, n essentially characterizes the evidence count, reflecting the effective number of visits to the specific state-action pair (s, a) . Applying the Law of Total Variance to this empirical estimator allows

for the theoretical decoupling of the total value uncertainty $\text{Var}(Q)$ into two distinct components:

$$\text{Var}[Q(s, a)] \approx \underbrace{\text{Var}_\theta[Q_\theta(s, a)]}_{\text{Epistemic}} + \underbrace{\frac{1}{n(s, a)} \mathbb{E}_\theta[\text{Var}[Z_\theta(s, a)]]}_{\text{Aleatoric uncertainty of } Z}, \quad (2)$$

$\text{Aleatoric uncertainty of } Q$

These components originate from distinct sources: *epistemic uncertainty* reflects structural disagreement among ensemble estimators due to diverse network initializations, whereas *aleatoric uncertainty* stems from inherent environmental and policy stochasticity. Crucially, the scaling factor $1/n$ dictates that the impact of the irreducible return uncertainty $\text{Var}(Z_\theta)$ on the value estimator is strictly modulated by the effective evidence count n .

This relationship reveals the origin of the *aleatoric impulse*. During early training, highly stochastic exploratory policies maximize return uncertainty, while severe data scarcity minimizes n . This combination sustains $1/n$ at a critically high level, directly amplifying intrinsic noise into a massive, transient surge in total value uncertainty. Through the maximization operator, this statistical impulse induces a severe overestimation bias, misleading the agent toward suboptimal actions and fundamentally derailing the learning trajectory.

3 Direct Evidential Regression

To bypass heuristic schedules common in traditional evidential regression, we propose a streamlined and principled framework termed **Direct Evidential Regression (DEAR)**. In this section, we establish this framework within a general parameter estimation setting, directly demonstrating its universality.

3.1 Methodology

Let x_θ denote a general parameter estimator parameterized by an online neural network with weights θ . To capture the aleatoric uncertainty inherent in the estimation process, we model x_θ as a Gaussian random variable, $x_\theta \sim \mathcal{N}(\mu_x, \sigma_x^2)$.

Higher-order mean. To approximate the latent mean μ_x , we introduce a target network parameterized by $\bar{\theta}$. This network serves as a stable empirical anchor and is maintained by tracking the online weights θ via an Exponential Moving Average (EMA): $\bar{\theta} \leftarrow \tau\theta + (1 - \tau)\bar{\theta}$, where $\tau \in (0, 1)$ denotes the smoothing coefficient.

Higher-order variance. Concurrently, an auxiliary prediction head $\sigma_{x,\theta}$ is appended to the online network to parameterize the variance σ_x^2 . It is optimized using a Gaussian Negative Log-Likelihood (NLL) objective:

$$\mathcal{L}(\theta) = \frac{(\text{sg}[x_\theta] - x_{\bar{\theta}})^2}{2\sigma_{x,\theta}^2} + \frac{1}{2} \log \sigma_x^2, \quad (3)$$

where $\text{sg}[\cdot]$ denotes the stop-gradient operation, ensuring that the gradient is not backpropagated through the online estimator x_θ in this objective.

3.2 Toy Experiment

To make the role of Direct Evidential Regression (DEAR) more transparent, we conduct a small-scale toy regression experiment as a mechanistic and visual complement to the full reinforcement learning results. This experiment is not intended to replace the MuJoCo and DMC benchmarks; instead, it isolates the uncertainty decomposition behavior of DEAR in a controlled setting, so that we can directly inspect whether the learned uncertainty rises in data-sparse regions and whether the fitting trajectory remains stable during training.

Connection to reinforcement learning. Although the toy task is one-dimensional, it captures the core ingredients that matter for critic learning in off-policy reinforcement learning. The nonlinear target function plays the role of a local slice of the action-value landscape, while the gradual expansion of observed samples mimics the fact that replay support grows over training. Under this analogy, the center region corresponds to frequently visited state-action pairs, whereas the outer region behaves

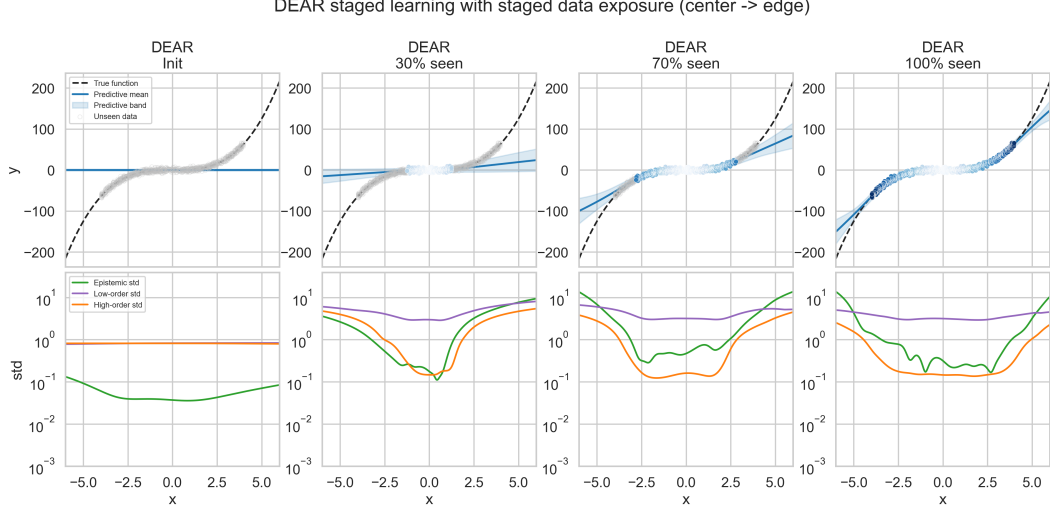


Figure 1: Training-time visualization of DEAR under staged data exposure. DEAR gradually extends the fitted curve from the data-rich center to the sparse outer region while maintaining a smoother uncertainty profile.

like sparsely covered or out-of-distribution (OOD) inputs. Therefore, a desirable uncertainty estimator should fit accurately in well-covered regions and increase uncertainty near poorly supported regions.

Experimental setup. We generate noisy observations from

$$y = x^3 + \epsilon, \quad \epsilon \sim \mathcal{N}(0, 3^2), \quad (4)$$

with training inputs sampled from $x \in [-4, 4]$ and evaluation conducted on the wider interval $[-6, 6]$ in order to explicitly expose OOD behavior. To mimic the progressive expansion of replay coverage in reinforcement learning, the training data are revealed in a staged manner from the center to the edge according to 30%, 70%, and 100% of the full dataset. The three stages are trained for 150, 150, and 200 epochs, respectively.

For DEAR, we use a 5-member ensemble, where each member predicts a mean together with two aleatoric components: a low-order term and a higher-order term. The ensemble variance is used as epistemic uncertainty, and the total predictive uncertainty is computed as the sum of epistemic, low-order aleatoric, and higher-order aleatoric terms. As a baseline, we additionally implement Deep Evidential Regression (DER) with the standard Normal-Inverse-Gamma parameterization and evidential regularization. For a fair comparison, DER is also evaluated with 5 random seeds and summarized with mean and standard deviation.

Training-time visualization. Figure 1 and Figure 2 visualize the staged learning process of DEAR and DER. In each plot, darker blue points indicate more recently observed samples. This design lets us simultaneously inspect the fitting trajectory, the order of data exposure, and the evolution of uncertainty.

Two observations are especially important. First, as more edge samples are revealed, DEAR expands its fitted curve smoothly toward the true cubic trend, which is consistent with the desired behavior of a critic whose value support improves as the replay buffer grows. Second, the uncertainty in DEAR is clearly higher in regions with fewer observed samples, especially near the boundary and outside the training interval. This is exactly the qualitative pattern we expect if the method is capturing data-dependent unreliability rather than simply inflating uncertainty everywhere.

Final comparison. Figure 3 and Figure 4 show the final fitting result and the uncertainty decomposition after training. DEAR achieves a visibly tighter fit to the ground-truth function and exhibits a more structured decomposition: the low-order aleatoric term stays relatively stable in the covered region, while the higher-order term becomes more prominent where the model has less reliable

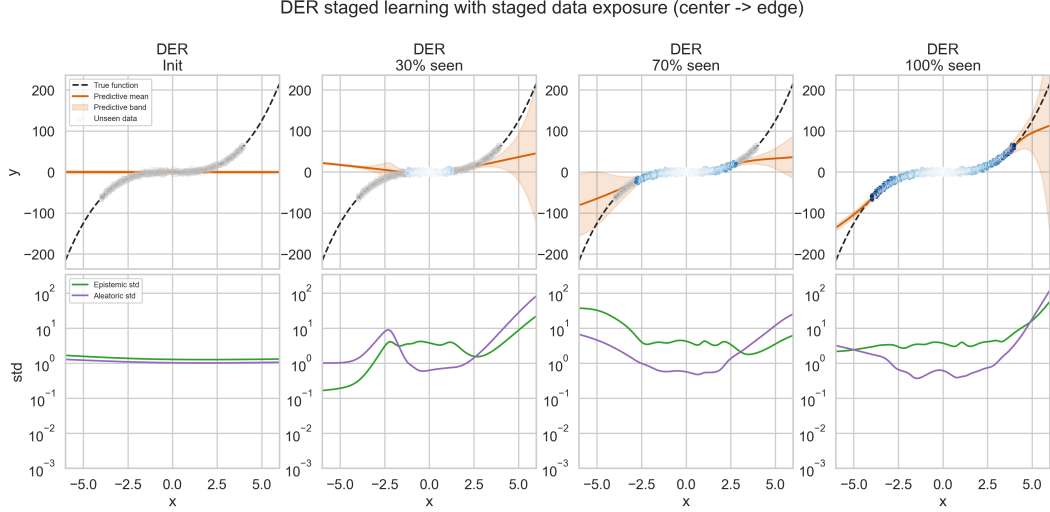


Figure 2: Training-time visualization of DER under staged data exposure. DER also reacts to unseen regions, but its uncertainty magnitude fluctuates more strongly across the training stages.

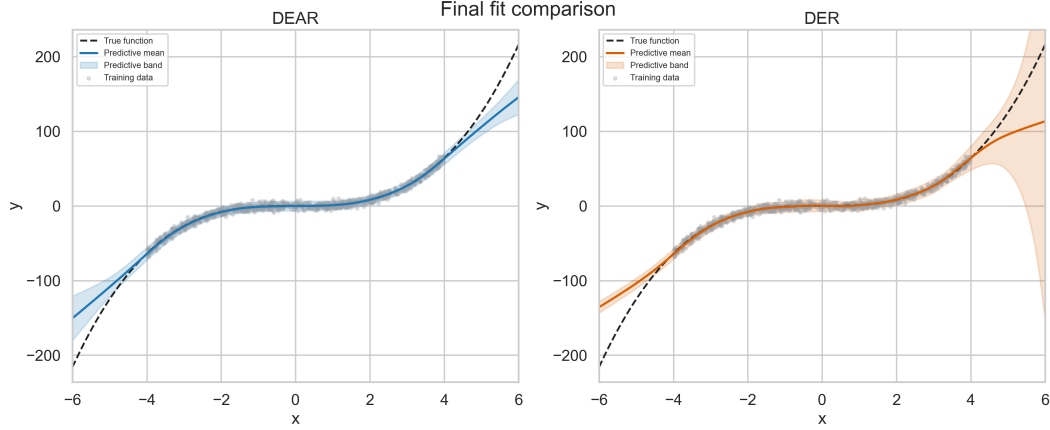


Figure 3: Final predictive fit comparison between DEAR and DER. DEAR yields lower fitting error over the full evaluation range.

support. In contrast, DER also raises uncertainty outside the training region, but the magnitude is considerably less stable.

Quantitative results. To reduce the influence of single-run randomness, we report 5-seed statistics in Table 1. DEAR achieves a lower RMSE than DER (18.93 ± 0.81 vs. 23.64 ± 2.24), indicating better overall fitting quality. More importantly, its in-domain total uncertainty remains compact (3.17 ± 0.07), while its OOD uncertainty rises to a moderate but consistent level (7.69 ± 1.18). DER, by contrast, exhibits much larger cross-seed variation in the OOD region (21.26 ± 20.04), suggesting that its evidence calibration is less stable in this setting.

Overall, the toy experiment supports the main motivation behind DEAR. The method not only provides a clearer decomposition of uncertainty, but also produces uncertainty estimates that are more aligned with data coverage: uncertainty remains controlled in well-observed regions and rises in sparse or OOD regions. This behavior is precisely what is needed when transferring DEAR to critic learning, because the resulting aleatoric estimate can be used downstream to damp unreliable target values and to explain why the full AIDER algorithm is able to suppress aleatoric impulse more robustly than heuristic evidential baselines.

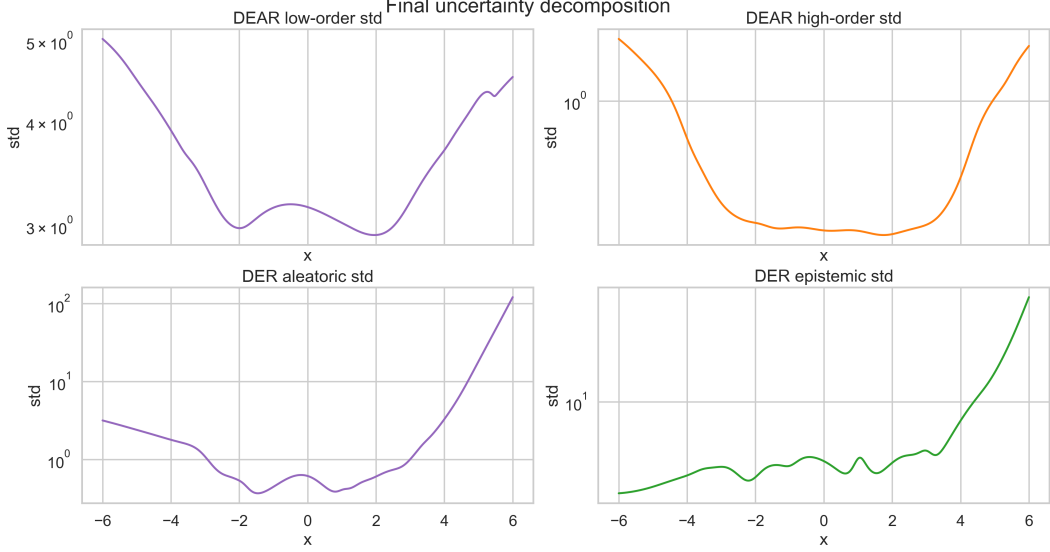


Figure 4: Final uncertainty decomposition comparison between DEAR and DER. DEAR shows a more stable uncertainty shape in the OOD region.

Table 1: Five-seed summary of the toy experiment. Lower RMSE is better.

Method	RMSE ↓	Mean total std ↓	In-domain std	OOD std
DEAR	18.93 ± 0.81	4.67 ± 0.39	3.17 ± 0.07	7.69 ± 1.18
DER	23.64 ± 2.24	9.60 ± 6.94	3.78 ± 0.39	21.26 ± 20.04

4 Aleatoric Impulse Damping via Direct Evidential Regression

While aleatoric impulse varies drastically across the state-action space depending on local visitations, existing damping mechanisms rely on global proxy variables, failing to capture this microscopic heterogeneity. To resolve this bottleneck, we propose **Aleatoric Impulse Damping via Direct Evidential Regression (AIDER)** by integrating DEAR.

4.1 Evidential Critic Augmentation

Within the DSAC architecture, we restructure the action-value function into an evidential neural network. Alongside the conventional modeling of the return distribution $Z \sim \mathcal{N}(Q_{\theta_i}, \sigma_{\text{ale-z}, \theta_i}^2)$, the i -th critic network is augmented with an additional output head. Instead of predicting the aleatoric value uncertainty directly, this head is designed to explicitly estimate the evidence count, $n_{\theta_i}(s, a)$. The aleatoric value uncertainty is subsequently parameterized through the structural relationship:

$$\sigma_{\text{ale-q}, \theta_i}^2(s, a) = \sigma_{\text{ale-z}, \theta_i}^2(s, a) / n_{\theta_i}(s, a). \quad (5)$$

Since evidence counts are stabler than aleatoric value uncertainty, this specific architectural design effectively mitigates the potentially disruptive impact of the DEAR loss gradients on the shared feature representation layers. Given that standard DSAC intrinsically relies on target networks parameterized by θ_i to stabilize value backups, this derived variance term $\sigma_{\text{ale-q}, \theta_i}^2$ is specifically designed to quantify the uncertainty of the target value Q_{θ_i} , rather than the online estimate Q_{θ_i} . Following DEAR, we maintain a higher-order target network $\hat{\theta}_i$ updated via EMA. Consequently, the target value estimator is explicitly modeled as a Gaussian distribution centered around this higher-order anchor: $Q_{\theta_i}(s, a) \sim \mathcal{N}(Q_{\hat{\theta}_i}(s, a), \sigma_{\text{ale-q}, \theta_i}^2(s, a))$.

4.2 Evidential Critic Learning

Following the proposed DEAR paradigm, we directly optimize this evidential parameterization by formulating it as a Negative Log-Likelihood (NLL) minimization problem. To empirically stabilize

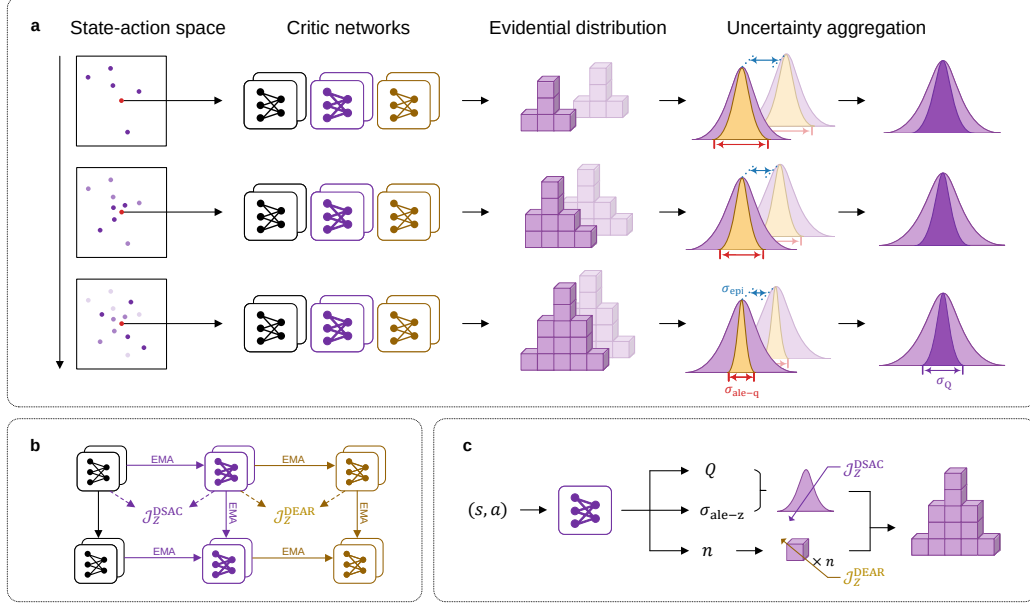


Figure 5: **The critic framework of DSAC-AIDER.** (a) **Temporal evolution:** Over training iterations (top to bottom), the accumulation of state-action visitations expands the evidence count n , driving the convergence of the discrete evidential distribution. Consequently, both epistemic uncertainty (σ_{epi}) and aleatoric value uncertainty ($\sigma_{\text{ale-q}}$) naturally diminish, narrowing the total value uncertainty. (b) **Optimization pipeline:** The underlying training workflow of the critic networks in (a). (c) **Single-pass parameterization:** The microscopic implementation of the network forward pass in (a).

the training process and prevent potential gradient explosion or variance collapse, the objective is scaled by the expectation of the learned variance, $\mathbb{E}[\sigma_{\text{ale-q}, \theta_i}^2]$. The new evidential loss for the i -th critic network is thus explicitly defined as:

$$\mathcal{J}_Z^{\text{DEAR}}(\theta_i) = \mathbb{E}_{\mathcal{D}}[\sigma_{\text{ale-q}, \theta_i}^2(s, a)] \cdot \mathbb{E}_{\mathcal{D}} \left[\left(\frac{(Q_{\theta_i}(s, a) - \hat{Q}_{\theta_i}(s, a))^2}{2\sigma_{\text{ale-q}, \theta_i}^2(s, a)} + \frac{1}{2} \log \sigma_{\text{ale-q}, \theta_i}^2(s, a) \right) \right] \quad (6)$$

This pure end-to-end mechanism drives the critic to autonomously calibrate the aleatoric value uncertainty $\sigma_{\text{ale-q}, \theta_i}^2(s, a)$ based strictly on the temporal consistency of its predictions. Consequently, this evidential objective seamlessly integrates with the base distributional reinforcement learning framework. The total loss for optimizing the critic network is simply formulated as the summation of the original value regression loss and the newly introduced evidential loss:

$$\mathcal{J}_Z^{\text{DSAC-AIDER}}(\theta_i) = \mathcal{J}_Z^{\text{DSAC-AID}}(\theta_i) + \mathcal{J}_Z^{\text{DEAR}}(\theta_i) \quad (7)$$

4.3 Uncertainty Utilization

Once the explicitly parameterized aleatoric value uncertainty is learned, we can accurately quantify the total value uncertainty without relying on any heuristic scaling. Specifically, we first compute the expected aleatoric value variance across the critic ensemble as $\sigma_{\text{ale-q}}^2(s, a) = \mathbb{E}_{\theta_i}[\sigma_{\text{ale-q}, \theta_i}^2(s, a)]$. The total value uncertainty $\sigma_Q(s, a)$ is then rigorously constructed by combining this directly learned aleatoric component with the epistemic uncertainty derived from the ensemble disagreement:

$$\sigma_Q(s, a) = \sqrt{\sigma_{\text{epi}}^2(s, a) + \sigma_{\text{ale-q}}^2(s, a)} \quad (8)$$

With $\sigma_Q(s, a)$ explicitly derived, its subsequent integration strictly follows the original AID mechanism. It gracefully substitutes the heuristically scaled uncertainty component of the previous generation, serving directly as the core penalty term to construct the pessimistic target bound for stable value evaluation, while concurrently acting as the dynamic exploration bonus for the decoupled

behavioral policy. This complete integration preserves the robust overestimation suppression and efficient exploration capabilities of the original architecture, yet entirely eliminates the complex closed-loop annealing schedules and hyperparameter sensitivity.

4.4 Pessimistic Evaluation

The critic maintains an ensemble of m evidential networks $\{\theta_i\}_{i=1}^m$, each modeling the return distribution as $Z_{\theta_i} \sim \mathcal{N}(Q_{\theta_i}, \sigma_{\text{ale-z}, \theta_i}^2)$, where $\sigma_{\text{ale-z}, \theta_i}^2$ captures the inherent aleatoric return uncertainty. For brevity in the gradient analysis below, we denote $\sigma_{\theta_i}^2 \equiv \sigma_{\text{ale-z}, \theta_i}^2$. Leveraging this ensemble, we derive the aggregated mean value from the target networks as: $Q(s, a) = \frac{1}{m} \sum_{i=1}^m Q_{\bar{\theta}_i}(s, a)$. The epistemic uncertainty is isolated from the ensemble disagreement as:

$$\sigma_{\text{epi}}^2(s, a) = \frac{1}{m-1} \sum_{i=1}^m [Q_{\bar{\theta}_i}(s, a) - Q(s, a)]^2. \quad (9)$$

Based on this decoupled metric together with the evidential aleatoric value uncertainty $\sigma_{\text{ale-q}}^2$, we construct a pessimistic target and implement a gradient modulation scheme to stabilize learning.

Pessimistic target. Aligning with our theory, we construct a lower confidence bound based on total value uncertainty to rectify the overestimation bias. This formulation explicitly integrates the aleatoric component with the epistemic one to effectively damp the aleatoric impulse. The enhanced critic target is defined as

$$y_q = r + \gamma \left[Q(s', a') - \lambda_{\text{LB}} \sigma_Q(s', a') - \alpha \log \pi_{\bar{\phi}}(a' | s') \right], \quad (10)$$

where the confidence coefficient λ_{LB} provides controllable pessimism. Here $\sigma_Q(s', a')$ incorporates the evidential aleatoric value uncertainty in place of any heuristic scaling. Under the specific setting of $\lambda_{\text{LB}} = 1/\sqrt{2}$ and $m = 2$, the above total uncertainty recovers the standard CDQ formulation as a special case.

Gradient modulation. The optimization of distributional value estimation inherently scales gradients by the inverse of the learned variance $\sigma_{\theta_i}^2$. In standard approaches like DSAC-T, the reliance on a shared target compels the learned variance to capture the total return uncertainty, which inherently entangles both aleatoric and epistemic components. To ensure the model learns a clean aleatoric distribution, σ_{θ_i} must track an independent sub-target y_{z_i} :

$$y_{z_i} = r + \gamma (z'_{\bar{\theta}_i} - \alpha \log \pi_{\bar{\phi}}(a' | s')) \Big|_{z'_{\bar{\theta}_i} \sim Z_{\bar{\theta}_i}(\cdot | s', a')}. \quad (11)$$

Since this target limits σ_{θ_i} to capture only aleatoric uncertainty, the standard inverse-variance scaling fails to dampen value updates in the presence of epistemic uncertainty. To enforce conservatism against out-of-distribution samples, we explicitly incorporate the epistemic uncertainty into the gradient scaler for the mean Q_{θ_i} . The resulting modulated gradients are formulated as:

$$\nabla_{\theta_i} J_Z \approx \mathbb{E}_{\mathcal{B}} \left[-\frac{\mathbb{E}_{\mathcal{B}}[\sigma_{\theta_i}^2 + \sigma_{\text{epi}}^2]}{\sigma_{\theta_i}^2 + \sigma_{\text{epi}}^2} (y_q - Q_{\theta_i}) \nabla Q_{\theta_i} - \frac{\mathbb{E}_{\mathcal{B}}[\sigma_{\theta_i}^2]}{\sigma_{\theta_i}^2} \frac{(y_{z_i} - Q_{\theta_i})^2 - \sigma_{\theta_i}^2}{\sigma_{\theta_i}} \nabla \sigma_{\theta_i} \right], \quad (12)$$

where $\sigma_{\theta_i}^2 + \sigma_{\text{epi}}^2$ estimates the total uncertainty of Z_{θ_i} . In contrast, the update for σ_{θ_i} remains scaled only by its own output, ensuring that the learning of aleatoric uncertainty is not hindered by epistemic uncertainty, allowing the model to swiftly capture the true noise distribution even in early training stages.

4.5 Optimistic Exploration

To empower the critic with the sufficient pessimism necessary to completely eliminate the aleatoric impulse, the accompanying risk of excessive underestimation must be proactively neutralized. Otherwise, it will inevitably force the policy to retreat into low-uncertainty action zones, abandoning high-potential regions. To counterbalance this, we decouple the actor into a target policy π_T for stable value backup and a behavior policy π_B for active exploration, parameterized by ϕ_T and ϕ_B .

For π_T , we aim to achieve risk neutrality with respect to inherent stochasticity while ensuring robustness against model ambiguity. Thus, π_T maximizes the Q-value penalized solely by epistemic uncertainty:

$$J_{\pi_T}(\phi_T) = \mathbb{E}_{s \sim \mathcal{B}, a \sim \pi_T} [Q(s, a) - \lambda_{\text{LB}} \sigma_{\text{epi}}(s, a) - \alpha \log \pi_{\phi_T}(a | s)]. \quad (13)$$

In contrast, π_B is designed to strictly counterbalance the critic’s suppression. It incorporates an upper confidence bound based on the total value uncertainty:

$$J_{\pi_B}(\phi_B) = \mathbb{E}_{s \sim \mathcal{B}, a \sim \pi_B} [Q(s, a) + \lambda_{\text{UB}} \sigma_Q(s, a) - \alpha \log \pi_{\phi_B}(a | s)], \quad (14)$$

where the confidence coefficient λ_{UB} controls optimism. Crucially, since σ_Q inherently contains the learned evidential aleatoric value uncertainty $\sigma_{\text{ale-q}}^2$, this optimism symmetrically counterbalances the critic’s pessimism, ensuring precise compensation for the aleatoric suppression.

5 Experiments

In this section, we conduct a comprehensive empirical evaluation of DSAC-AIDER across two distinct domains. First, we benchmark against state-of-the-art algorithms on standard high-dimensional continuous control tasks (Section 5.2), establishing the general efficacy of our method. Subsequently, we deploy DSAC-AIDER on a practical four-wheel-drive vehicle longitudinal tracking task (Section 5.3), demonstrating its applicability to real-world engineering problems. This two-tiered structure—general benchmark followed by specific application—provides a holistic assessment of DSAC-AIDER’s capabilities.

5.1 Experimental Setup

Baselines. We evaluate DSAC-AIDER against ten representative algorithms categorized into two paradigms: (1) model-free baselines, including classic PPO [19] and SAC [20], state-of-the-art distributional methods DSAC-T [8], DoubleGum [11], and BRO [10], as well as the most closely related predecessor DSAC-AID [7]; and (2) model-based baselines, comprising state-of-the-art off-policy approaches DreamerV3 [21], TD-MPC2 [22], BMPC [23], and BOOM [24]. Including DSAC-AID as a baseline isolates the specific contribution of the DEAR operator and the direct evidential uncertainty estimation over the original heuristic annealing mechanism.

Benchmarks. We evaluate on two distinct types of environments. For the standard benchmark (Section 5.2), DSAC-AIDER is tested on a challenging suite of 10 high-dimensional locomotion tasks drawn from MuJoCo [25] continuous control suite in OpenAI Gym [26] and the DeepMind Control Suite (DMC) [27]. The selected 3 Gym-MuJoCo tasks and 7 DMC tasks feature three of the most complex agents, including humanoid, ant, and dog. These tasks demand sophisticated balance and coordination in high-dimensional observation and continuous action spaces. For the vehicle control application (Section 5.3), we construct a custom four-wheel-drive longitudinal dynamics environment, whose specifications are described therein.

5.2 Comparative Evaluation on Standard Benchmarks

We first evaluate DSAC-AIDER against the competitive baselines on the standard MuJoCo and DMC locomotion benchmarks. This section reports the head-to-head comparison across all 10 tasks, including training curves and the corresponding total average return (TAR) metrics.

Results on Gym-MuJoCo. On the two Gym-MuJoCo tasks, DSAC-AIDER shows competitive performance against strong model-free baselines. On Humanoid-v3, DSAC-AIDER achieves the best TAR of $12.177 \pm 0.563\text{k}$, outperforming DSAC-AID by approximately 4.6% and improving over DSAC-T by approximately 12.5%. This result indicates that the proposed direct evidential uncertainty estimation is beneficial for high-dimensional humanoid locomotion, where unstable value estimation can easily degrade policy learning. On Ant-v3, DSAC-AIDER obtains $8.300 \pm 0.073\text{k}$, ranking second behind DSAC-AID, while still outperforming PPO, SAC, DSAC-T, BRO, and DoubleGum. The normalized average score on the two Gym-MuJoCo tasks is 0.821 ± 0.029 for DSAC-AIDER, which remains competitive but is lower than DSAC-AID’s 0.890 ± 0.118 .

Results on DMC Suite. On the four DMC Suite tasks, DSAC-AIDER achieves the best normalized average performance among all model-free algorithms, obtaining 0.871 ± 0.137 compared with

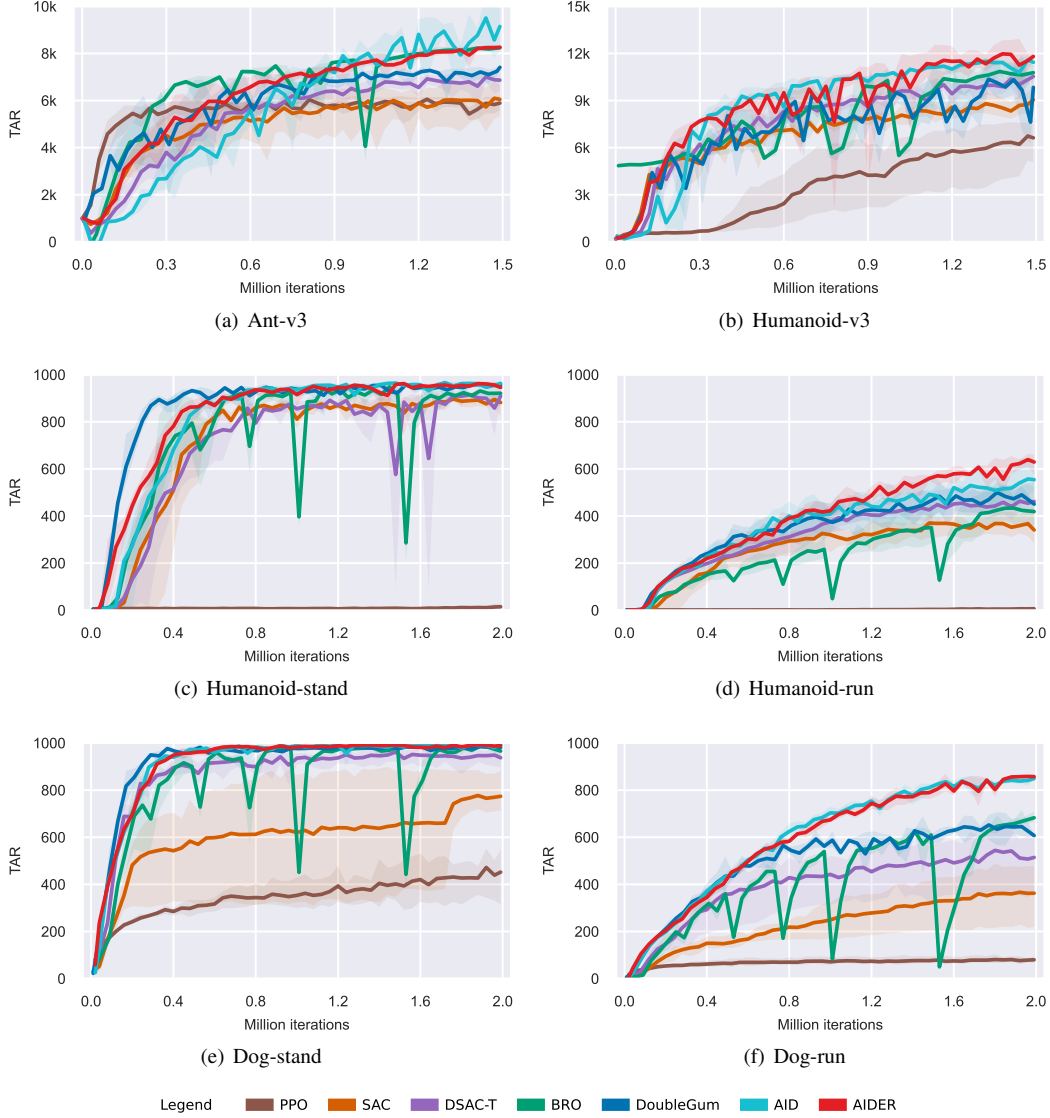


Figure 6: **Training curves of model-free algorithms on standard benchmarks.** The solid lines represent the mean, while the shaded regions indicate the confidence interval over multiple seeds.

0.853 ± 0.170 for DSAC-AID and 0.796 ± 0.197 for DoubleGum. At the task level, DSAC-AIDER performs especially well on dynamic locomotion tasks. On Dog-run, it achieves the best TAR of 870.6 ± 5.1 , slightly exceeding DSAC-AID and substantially improving over DSAC-T. On Humanoid-run, DSAC-AIDER obtains 648.1 ± 22.9 , outperforming DSAC-AID by approximately 12.7% and DSAC-T by approximately 37.0%. For standing tasks, DSAC-AIDER remains highly competitive, ranking second on both Dog-stand and Humanoid-stand with scores of 995.7 ± 2.0 and 969.5 ± 3.2 , respectively. These results indicate that DSAC-AIDER is particularly advantageous in more dynamic and unstable locomotion scenarios, where accurate uncertainty estimation is critical for suppressing value overestimation and maintaining stable exploration.

Comparison with DSAC-AID. Compared with its predecessor DSAC-AID, DSAC-AIDER replaces the heuristic uncertainty scaling mechanism with the DEAR-based direct estimation of aleatoric value uncertainty. The empirical results show that this modification brings clear gains on several challenging dynamic tasks. In particular, DSAC-AIDER improves over DSAC-AID on Humanoid-v3, Dog-run, and Humanoid-run, and achieves a higher normalized average score on the four DMC Suite tasks. The improvement is most pronounced on Humanoid-run, where DSAC-AIDER increases TAR

Table 2: Total Average Return (TAR) of model-free algorithms on DMC Suite and Gym-MuJoCo tasks. Mean $\pm$ Std over multiple seeds. Gym-MuJoCo values are in **k** (10^3). **Darker Blue** = best, **Lighter Blue** = second-best. The average scores of individual benchmarks ($\dagger$) are normalized by task-specific thresholds before averaging.

Task	PPO	SAC	DSAC-T	BRO	DoubleGum	DSAC-AID	DSAC-AIDER (ours)
Ant-v3 (k)	6.213 $\pm$ 0.151	6.433 $\pm$ 0.806	7.086 $\pm$ 0.260	8.251 $\pm$ 0.057	7.475 $\pm$ 0.021	10.029 $\pm$ 0.440	8.300 $\pm$ 0.073
Humanoid-v3 (k)	6.869 $\pm$ 1.563	9.389 $\pm$ 0.646	10.829 $\pm$ 0.243	10.974 $\pm$ 0.345	10.849 $\pm$ 0.850	11.643 $\pm$ 0.218	12.177 $\pm$ 0.563
AVG. Gym-MuJoCo†	0.540 $\pm$ 0.111	0.635 $\pm$ 0.065	0.715 $\pm$ 0.023	0.778 $\pm$ 0.050	0.735 $\pm$ 0.042	0.890 $\pm$ 0.118	0.821 $\pm$ 0.029
Dog-stand	479.3 $\pm$ 85.6	790.4 $\pm$ 77.3	966.5 $\pm$ 25.4	992.5 $\pm$ 2.3	992.7 $\pm$ 1.8	996.9 $\pm$ 0.1	995.7 $\pm$ 2.0
Dog-run	85.6 $\pm$ 13.8	372.4 $\pm$ 108.0	558.0 $\pm$ 124.9	696.0 $\pm$ 20.7	699.6 $\pm$ 37.1	868.2 $\pm$ 6.0	870.6 $\pm$ 5.1
Humanoid-stand	15.6 $\pm$ 9.3	907.5 $\pm$ 15.9	933.3 $\pm$ 2.0	961.5 $\pm$ 5.7	966.7 $\pm$ 3.2	973.0 $\pm$ 0.3	969.5 $\pm$ 3.2
Humanoid-run	6.8 $\pm$ 3.7	394.9 $\pm$ 46.9	473.0 $\pm$ 49.0	475.3 $\pm$ 51.9	523.6 $\pm$ 42.1	575.0 $\pm$ 56.7	648.1 $\pm$ 22.9
AVG. DMC Suite†	0.147 $\pm$ 0.199	0.616 $\pm$ 0.247	0.733 $\pm$ 0.230	0.781 $\pm$ 0.213	0.796 $\pm$ 0.197	0.853 $\pm$ 0.170	0.871 $\pm$ 0.137

from 575.0 ± 56.7 to 648.1 ± 22.9 , suggesting that directly learned value uncertainty provides a more reliable penalty and exploration signal in highly unstable locomotion tasks. Meanwhile, DSAC-AID still performs slightly better on Ant-v3, Dog-stand, Humanoid-stand, and the Gym-MuJoCo average. Overall, these results suggest that AIDER does not uniformly dominate the previous AID mechanism on every benchmark, but it improves performance and stability on the more challenging dynamic locomotion tasks and leads to the best average performance on the DMC Suite benchmark.

5.3 Application to Vehicle Longitudinal Control

To further validate the practical utility of DSAC-AIDER beyond simulated benchmarks, we deploy the algorithm on a custom four-wheel-drive (4WD) longitudinal vehicle dynamics environment. Unlike the standard locomotion tasks where the agent learns to coordinate limb movements, this task requires precise continuous control of wheel torques to track a time-varying reference speed, representing a distinct category of engineering-relevant control problems.

5.3.1 Task Description

The environment is built upon the GOPS framework [28] and models a vehicle with independent torque control at each of the four wheels. The state space comprises the longitudinal velocity v , acceleration a_x , velocity tracking error v_{err} , the four wheel torques $T_{\text{fl}}, T_{\text{fr}}, T_{\text{rl}}, T_{\text{rr}}$, and a 10-step preview horizon of the reference speed, providing the agent with look-ahead information of the target trajectory. The action space is four-dimensional, representing the commanded motor torque for each wheel, bounded within $[-1500, +2000]$ N·m. The control operates at 100 Hz ($\Delta t = 0.05$ s) and each episode spans 1000 steps. Experiments are conducted on an AutoDL cloud platform equipped with an RTX 3090 GPU, Docker containerization, and CUDA 12.1.

The reference speed profile is sinusoidal with a mean of 20 m/s and an amplitude of 3.0 m/s, yielding a feasible tracking range of 17–23 m/s. This design simulates realistic driving scenarios with continuous speed variations. To avoid the sparse-penalty collapse observed in early training iterations, the reward function is carefully shaped with three components: (i) a *direction-matching bonus* that encourages the correct sign of the total torque relative to the velocity error; (ii) a *speed-matching bonus* that grants +200 when $|v_{\text{err}}| < 1$ m/s; and (iii) *soft penalties* for tracking error magnitude and control jerk. The total reward is scaled by 0.1 to stabilize the distributional Q-value targets.

5.3.2 Training Dynamics

Figure 7 presents the training curves of DSAC-AIDER on the Veh4WD tracking task. The training converges at approximately 100k iterations. The mean Q-value rises from an initial negative estimate to approximately +1671, indicating that the critic successfully learns a positive value landscape. The Total Average Return (TAR) stabilizes around 416,333 after 50k steps. Concurrently, the policy

Table 3: Tracking performance of DSAC-AIDER on the Veh4WD task, evaluated over 5 independent test episodes (2000 steps each).

Metric	Value
TAR	416,333
RMSE (m/s)	0.349
MAE (m/s)	0.272
Tracking Accuracy	$\sim 1.5\%$
Policy Std (final)	0.60
Episode Completion	5/5 (100%)

standard deviation decays from an exploratory level of 1.65 to a near-deterministic 0.60, reflecting the agent’s progressive transition from exploration to exploitation as tracking proficiency improves.

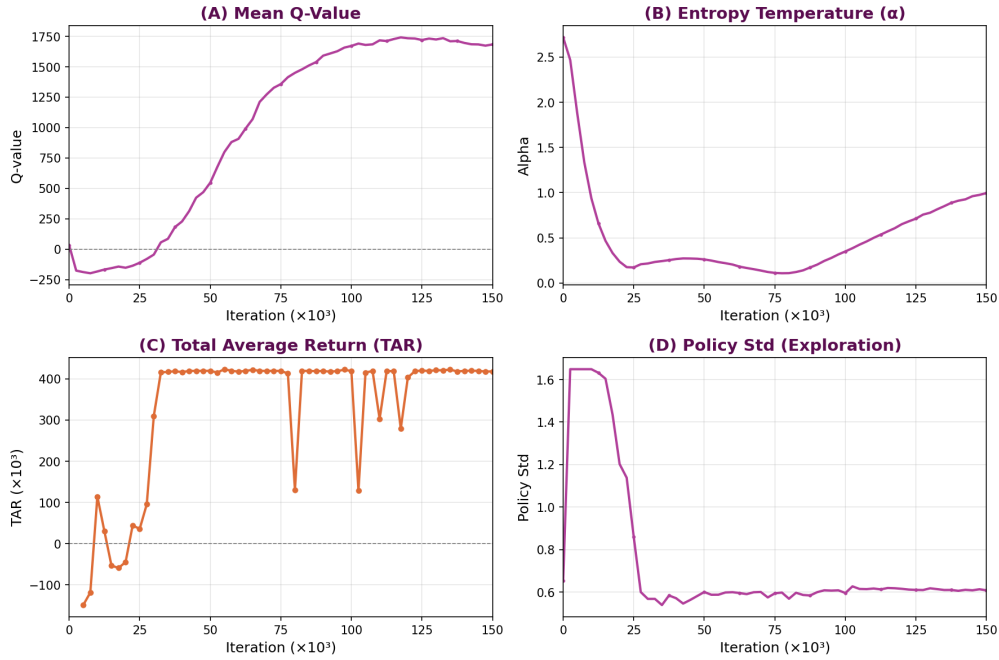


Figure 7: **Training curves of DSAC-AIDER on the Veh4WD longitudinal tracking task (first 150k steps).** (A) Mean Q-value, rising from negative to approximately +1671. (B) Entropy temperature α , automatically tuned to balance exploration and exploitation. (C) Total Average Return (TAR), stabilizing around 416,333 after 50k steps. (D) Policy standard deviation (exploration noise), decaying from 1.65 to 0.60.

5.3.3 Tracking Performance

We evaluate the trained policy over five independent episodes, each spanning 2000 steps. All five episodes complete the full horizon without premature termination. Table 3 summarizes the quantitative tracking metrics. The policy achieves a mean Root Mean Square Error (RMSE) of 0.349 m/s and a mean Mean Absolute Error (MAE) of 0.272 m/s, corresponding to a tracking accuracy of approximately 1.5% relative to the target speed range (17–23 m/s). These results demonstrate that DSAC-AIDER reliably maintains accurate speed tracking across varying reference velocities.

Figure 8 visualizes the tracking behavior during a representative test episode (TAR = 415,752.5). The top panel shows the actual vehicle speed closely following the sinusoidal reference, remaining within the ± 1 m/s tolerance band for the vast majority of the trajectory. The middle panel displays the four-wheel torque commands, which exhibit coordinated behavior to maintain tracking. The bottom

panel presents the per-step reward and longitudinal acceleration a_x , demonstrating consistent reward accumulation throughout the episode.

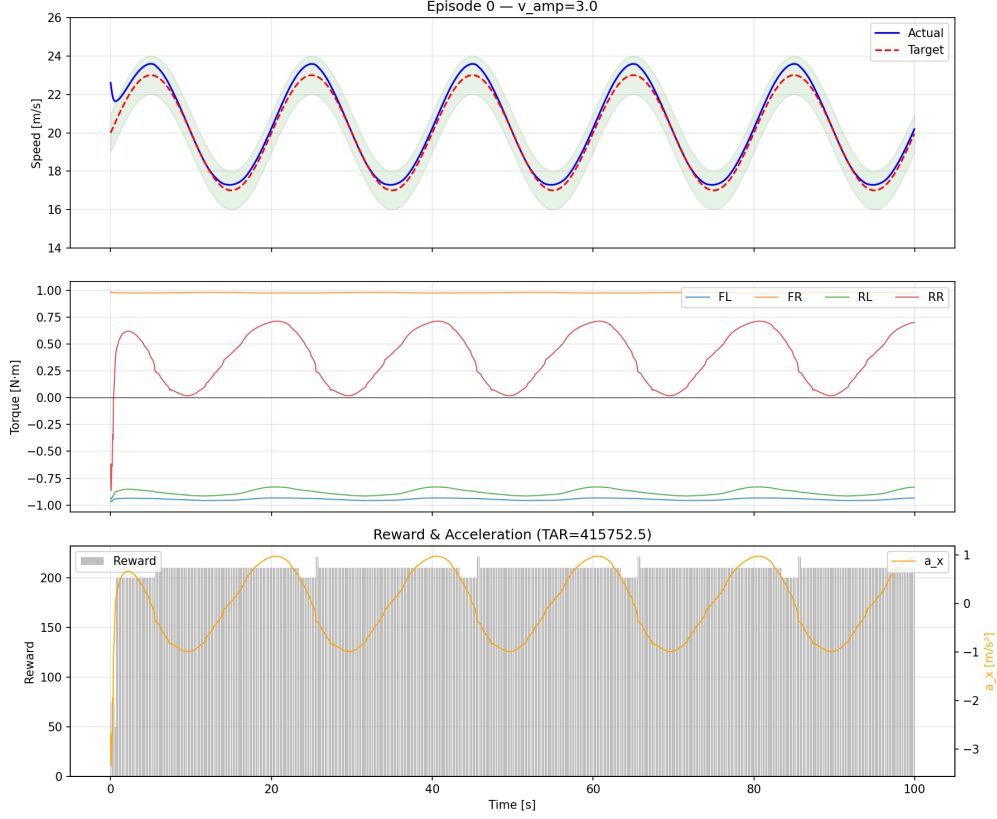


Figure 8: **Simulated speed tracking performance of DSAC-AIDER on a representative test episode ($v_{\text{amp}} = 3.0$).** **Top panel:** Actual vehicle speed (blue) vs. reference target speed (red dashed), with the ± 1 m/s tolerance band shaded in green. **Middle panel:** Four-wheel torque commands (T_{fl} , T_{fr} , T_{rl} , T_{rr}). **Bottom panel:** Per-step reward (gray bars) and longitudinal acceleration a_x (orange curve). TAR = 415,752.5.

Discussion. A notable observation is that the learned policy does not enforce torque symmetry between the left and right wheels; the net torque imbalance suggests that the agent exploits the longitudinal model’s lack of yaw dynamics to maintain tracking. While this does not degrade the reported tracking accuracy, it indicates that the current reward formulation does not penalize physically implausible torque distributions. Future iterations could incorporate a symmetry regularization term into the reward function or reduce the action space to a physically constrained torque distribution to encourage more realistic control strategies. Overall, the vehicle control results demonstrate that DSAC-AIDER effectively transfers from standard simulated benchmarks to a practical engineering domain, maintaining stable learning dynamics and accurate tracking performance.

6 Related Work

This section reviews the literature from two synergistic perspectives: aleatoric uncertainty quantification and value overestimation suppression.

Aleatoric uncertainty quantification. Existing methods follow two primary paradigms: (1) *Distributional regression*. This paradigm models the entire return distribution. Existing methods generally fall into two categories: quantile-based approaches [29, 30], which approximate arbitrary distributions through a discrete set of quantiles, and Gaussian-based approaches [8, 7], which parameterize the distribution via mean and variance for analytic efficiency. While these methods explicitly represent

return uncertainty, they typically treat the distribution parameters as point estimates, failing to quantify the high-order estimation uncertainty inherent in the learning process. (2) *Evidential regression*. Evidential regression [15] parameterizes higher-order distributions within a single sampling-free forward pass. However, traditional evidential frameworks rely on heuristic regularizing losses to penalize evidence, which often fails to accurately represent the true underlying uncertainty in non-stationary prediction. Our DEAR operator addresses this limitation by adopting a target network as an empirical anchor, bypassing heuristic evidence-curbing in favor of a more grounded and precise estimation of high-order value uncertainty.

Value overestimation suppression. Approaches for mitigating value overestimation in off-policy RL primarily follow two paradigms: network and objective regularization. (1) *Network regularization*. This paradigm stabilizes approximation via specific architectural constraints, exemplified by LayerNorm [31] in BRO [10], GroupNorm [32] in DoubleGum [11] and hyperspherical normalization in Simba [33]. Despite their stability benefits, these static structural priors often impose a constant regularization overhead that may inadvertently restrict model expressivity in low-risk regions. (2) *Objective regularization*. This paradigm mitigates overestimation by augmenting the learning objective with uncertainty constraints or penalties. Implicit approaches leverage structural heuristics or distributional properties without direct penalty terms, such as TD3 [34] and DSAC-T [8]. Explicit approaches, conversely, construct quantifiable penalties based on specific uncertainty sources. While traditional methods often target isolated sources [35, 36, 37], recent frameworks like DoubleGum [11] incorporate both epistemic and aleatoric components. However, DoubleGum targets the total return uncertainty, which may not perfectly align with the estimation bias of the value function. In contrast, AID [7] and AIDER explicitly target the precise total value uncertainty, providing a more direct and theoretically grounded suppression of overestimation.

7 Conclusion

In this paper, we address the aleatoric impulse—a transient yet destructive value overestimation surge in off-policy RL, driven by intrinsic estimation uncertainty. To resolve this, we propose DEAR, a generalized evidential regression operator that uses an EMA anchor to directly quantify higher-order aleatoric uncertainty end-to-end, bypassing heuristic approximations and rigid conjugate priors. Building on DEAR, we develop AIDER, which restructures the critic as an evidential network to explicitly learn aleatoric value uncertainty. This unified total uncertainty (aleatoric + epistemic) enables robust pessimistic penalization and dynamic exploration, while completely eliminating complex annealing schedules and indirect variance reconstruction. AIDER achieves state-of-the-art evaluation stability and efficient exploration across challenging continuous control benchmarks.

8 Team Roles and Responsibilities

To efficiently manage the complexities of theoretical analysis, algorithmic development, and empirical validation, the project is collaboratively distributed among our eight-member team. The specific responsibilities are allocated as follows:

- **Zhouyang Yu:** Serves as project co-coordinator. Primary responsibilities include the architectural design, core code implementation, and empirical validation of the fundamental AID framework, alongside overall team coordination.
- **Haoyuan Liu:** Serves as project co-coordinator. Focuses extensively on the hyperparameter tuning and algorithmic optimization of the AIDE framework, while assisting in managing team workflows.
- **Zhixin Ren:** Leads the theoretical analysis of approximation conditions. Specifically investigates the conditions under which the uncertainty decomposition in AID holds (analyzing over/under-estimation trends and mitigation tricks) and conducts the approximation analysis for the newly proposed evidential learning method in AIDE.
- **Kaidi Zha:** Explores the integration of the AID framework with Diffusion and Normalizing Flow models to extend its application to multi-modal policies.
- **Hao Fan:** Collaborates on the exploration of applying AID to multi-modal policies via Diffusion/Flow models.

- **Junlang Zhu:** In charge of the validation and empirical analysis of the evidential learning toy experiments.
- **Wuzhe Zhu:** Responsible for configuring and setting up the comprehensive benchmark testing environments. Additionally, handles the evaluation of trained policies and the rendering of high-quality simulation videos for visual verification of agent behaviors.
- **Wei Song:** Leads the practical downstream application of the AID mechanism to vehicle drive-brake cooperative control scenarios. Furthermore, conducts algorithmic tuning and performance validation for AID, and designs specialized evaluation metrics to accurately quantify value overestimation.

References

- [1] David Silver, Julian Schrittwieser, Karen Simonyan, Ioannis Antonoglou, Aja Huang, Arthur Guez, Thomas Hubert, Lucas Baker, Matthew Lai, Adrian Bolton, et al. Mastering the game of go without human knowledge. *Nature*, 550(7676):354–359, 2017.
- [2] Oriol Vinyals, Igor Babuschkin, Wojciech M Czarnecki, Michaël Mathieu, Andrew Dudzik, Junyoung Chung, David H Choi, Richard Powell, Timo Ewalds, Petko Georgiev, et al. Grandmaster level in starcraft ii using multi-agent reinforcement learning. *Nature*, 575(7782):350–354, 2019.
- [3] Nikita Rudin, David Hoeller, Philipp Reist, and Marco Hutter. Learning to walk in minutes using massively parallel deep reinforcement learning. In *Conference on robot learning*, pages 91–100. PMLR, 2022.
- [4] Jingliang Duan, Yangang Ren, Fawang Zhang, Jie Li, Shengbo Eben Li, Yang Guan, and Keqiang Li. Encoding distributional soft actor-critic for autonomous driving in multi-lane scenarios [research frontier]. *IEEE Computational Intelligence Magazine*, 19(2):96–112, 2024.
- [5] Sebastian Thrun and Anton Schwartz. Issues in using function approximation for reinforcement learning. In *Proceedings of the 1993 connectionist models summer school*, pages 255–263. Psychology Press, 2014.
- [6] Eyke Hüllermeier and Willem Waegeman. Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods. *Machine learning*, 110(3):457–506, 2021.
- [7] Zhouyang Yu, Guojian Zhan, Yang Guan, Jingliang Duan, Letian Tao, and Shengbo Eben Li. Taming the aleatoric impulse in off-policy reinforcement learning. In *International conference on machine learning*, 2026.
- [8] Jingliang Duan, Wenxuan Wang, Liming Xiao, Jiaxin Gao, Shengbo Eben Li, Chang Liu, Ya-Qin Zhang, Bo Cheng, and Keqiang Li. Distributional soft actor-critic with three refinements. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025.
- [9] Vincent Mai, Kaustubh Mani, and Liam Paull. Sample efficient deep reinforcement learning via uncertainty estimation. In *International conference on learning representations*, pages 10283–10306, 2022.
- [10] Michal Nauman, Mateusz Ostaszewski, Krzysztof Jankowski, Piotr Miłoś, and Marek Cygan. Bigger, regularized, optimistic: scaling for compute and sample efficient continuous control. *Advances in neural information processing systems*, 37:113038–113071, 2024.
- [11] David Yu-Tung Hui, Aaron C Courville, and Pierre-Luc Bacon. Double gumbel q-learning. *Advances in neural information processing systems*, 36:2580–2616, 2023.
- [12] Junyu Gao, Mengyuan Chen, Liangyu Xiang, and Changsheng Xu. A comprehensive survey on evidential deep learning and its applications. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025.
- [13] Murat Sensoy, Lance Kaplan, and Melih Kandemir. Evidential deep learning to quantify classification uncertainty. *Advances in neural information processing systems*, 31, 2018.

- [14] Bertrand Charpentier, Daniel Zügner, and Stephan Günnemann. Posterior network: Uncertainty estimation without ood samples via density-based pseudo-counts. *Advances in neural information processing systems*, 33:1356–1367, 2020.
- [15] Alexander Amini, Wilko Schwarting, Ava Soleimany, and Daniela Rus. Deep evidential regression. *Advances in neural information processing systems*, 33:14927–14937, 2020.
- [16] Nis Meinert, Jakob Gawlikowski, and Alexander Lavin. The unreasonable effectiveness of deep evidential regression. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 9134–9142, 2023.
- [17] Yuefei Wu, Bin Shi, Bo Dong, Qinghua Zheng, and Hua Wei. The evidence contraction issue in deep evidential regression: Discussion and solution. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 21726–21734, 2024.
- [18] Shengbo Eben Li. *Reinforcement learning for sequential decision and optimal control*. Springer, 2023.
- [19] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- [20] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *International conference on machine learning*, pages 1861–1870. Pmlr, 2018.
- [21] Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse control tasks through world models. *Nature*, pages 1–7, 2025.
- [22] Nicklas Hansen, Hao Su, and Xiaolong Wang. TD-MPC2: Scalable, robust world models for continuous control. In *International conference on learning representations*, pages 3534–3563, 2024.
- [23] Yuhang Wang, Hanwei Guo, Sizhe Wang, Long Qian, and Xuguang Lan. Bootstrapped model predictive control. In *International conference on learning representations*, pages 44283–44301, 2025.
- [24] Guojian Zhan, Likun Wang, Xiangteng Zhang, Jiaxin Gao, Masayoshi Tomizuka, and Shengbo Eben Li. Bootstrap off-policy with world model. In *Advances in neural information processing systems*, volume 39, 2025.
- [25] Emanuel Todorov, Tom Erez, and Yuval Tassa. Mujoco: A physics engine for model-based control. In *2012 IEEE/RSJ international conference on intelligent robots and systems*, pages 5026–5033. IEEE, 2012.
- [26] Greg Brockman, Vicki Cheung, Ludwig Pettersson, Jonas Schneider, John Schulman, Jie Tang, and Wojciech Zaremba. Openai gym. *arXiv preprint arXiv:1606.01540*, 2016.
- [27] Yuval Tassa, Yotam Doron, Alistair Muldal, Tom Erez, Yazhe Li, Diego de Las Casas, David Budden, Abbas Abdolmaleki, Josh Merel, Andrew Lefrancq, et al. Deepmind control suite. *arXiv preprint arXiv:1801.00690*, 2018.
- [28] Wenxuan Wang, Yuhang Zhang, Jiaxin Gao, Yuxuan Jiang, Yujie Yang, Zhilong Zheng, Wenjun Zou, Jie Li, Congsheng Zhang, Wenhan Cao, et al. Gops: A general optimal control problem solver for autonomous driving and industrial control applications. *Communications in Transportation Research*, 3:100096, 2023.
- [29] Will Dabney, Georg Ostrovski, David Silver, and Rémi Munos. Implicit quantile networks for distributional reinforcement learning. In *International conference on machine learning*, pages 1096–1105. PMLR, 2018.
- [30] Arsenii Kuznetsov, Pavel Shvechikov, Alexander Grishin, and Dmitry Vetrov. Controlling overestimation bias with truncated mixture of continuous distributional quantile critics. In *International conference on machine learning*, pages 5556–5566. PMLR, 2020.

- [31] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E Hinton. Layer normalization. *arXiv preprint arXiv:1607.06450*, 2016.
- [32] Yuxin Wu and Kaiming He. Group normalization. In *Proceedings of the European conference on computer vision (ECCV)*, pages 3–19, 2018.
- [33] Hojoon Lee, Youngdo Lee, Takuma Seno, Donghu Kim, Peter Stone, and Jaegul Choo. Hyperspherical normalization for scalable deep reinforcement learning. *arXiv preprint arXiv:2502.15280*, 2025.
- [34] Scott Fujimoto, Herke Hoof, and David Meger. Addressing function approximation error in actor-critic methods. In *International conference on machine learning*, pages 1587–1596. PMLR, 2018.
- [35] Sicen Li, Qinyun Tang, Yiming Pang, Xinmeng Ma, and Gang Wang. Realistic actor-critic: A framework for balance between value overestimation and underestimation. *Frontiers in Neurorobotics*, 16:1081242, 2023.
- [36] Edoardo Cetin and Oya Celiktutan. Learning pessimism for reinforcement learning. In *Proceedings of the AAAI conference on artificial intelligence*, volume 37, pages 6971–6979, 2023.
- [37] William R Clements, Bastien Van Delft, Benoît-Marie Robaglia, Reda Bahi Slaoui, and Sébastien Toth. Estimating risk and uncertainty in deep reinforcement learning. *arXiv preprint arXiv:1905.09638*, 2019.